LEARN / DATA CENTERS / BEGINNER EDITION

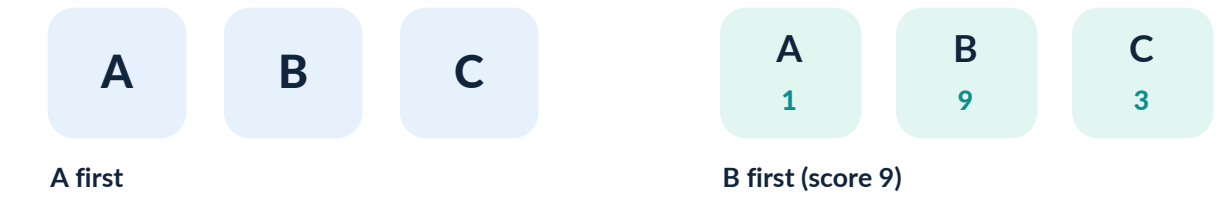
FIFO (First In, First Out) · GPU (Graphics Processing Unit)

QUEUES & PRIORITY HEAPS

A structure for waiting can determine who receives expensive hardware next.

SCAN / LEARN

THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 FIFO (FIRST IN, FIRST OUT)

A queue normally removes the oldest item first. This is fair by arrival order, but not always by urgency.

2 PRIORITY QUEUE

Returns the highest-priority item first. The queue can be implemented with a heap (a special tree shape).

3 HEAP (TREE FOR PRIORITY)

A heap lets software find a highest- or lowest-priority item quickly; it does not keep all items fully sorted.

4 THE TRADE-OFF

Strict priority can starve low-priority jobs. Real schedulers may add fairness, quotas or aging rules.

MEMORY HOOK

The scheduler is not “thinking”: it is following ordering and capacity rules.

DISCUSS WITHOUT COMPUTERS

Which structure chooses the oldest job, and which picks the most urgent?

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